

LECTURE-21

(Pre-treatment)

○ Chemical Pre-treatment

Chemical pretreatment methods are used more often than biological or physical pretreatment methods because they are more effective and enhance the biodegradation of complex materials. Common chemicals used in chemical pretreatment methods for improving the AD performance of agricultural residues are sulfuric acid (H_2SO_4), hydrochloric acid (HCl), acetic acid (CH_3COOH), sodium hydroxide (NaOH), potassium hydroxide (KOH), lime ($\text{Ca}(\text{OH})_2$), aqueous ammonia ($\text{NH}_3 \cdot \text{H}_2\text{O}$), and hydrogen peroxide (H_2O_2).

Alkali Pretreatment

- Alkali pretreatment is a widely studied chemical pretreatment method which is based on the solubilization of lignin in the alkali solution.
- The various alkaline reagents used commonly for alkali pretreatment are the hydroxides of sodium, potassium, calcium and ammonium.
- Among these, sodium hydroxide is the most effective alkali pretreatment.
- A saponification reaction takes place throughout the alkali pretreatment process which causes cleavage of the intermolecular ester linkages between hemicelluloses and lignin.
- This results in solubilisation of lignin and hemicellulose fragments in the alkali solution and brings the cellulose in the interaction of enzymes.
- Also, alkali pretreatment changes the lignocellulosic structure via cellulose swelling that leads to reduction in crystallinity and degree of polymerization thereby increasing internal surface area.
- In addition, removal of acetyl groups and uronic acid substitutions in hemicelluloses during alkali pretreatment also increases the accessibility of the carbohydrates to enzymatic hydrolysis.
- Various studies reflect many advantages of the alkali pretreatment technique.
- A study on the alkaline pretreatment of rice straw for biomethane production reported that 1% NaOH at room temperature for 3 h significantly reduces the hemicellulose and lignin contents while the cellulose content remains unaltered.
- This led to an increase of methane yield by more than 34% compared to untreated rice straw.
- Alkali pretreatment is an effective technique in removing lignin and makes carbohydrates more exposed to use for the downstream processes. However, a major disadvantage of the technique is the recovery of the added alkalis which requires further investigations.

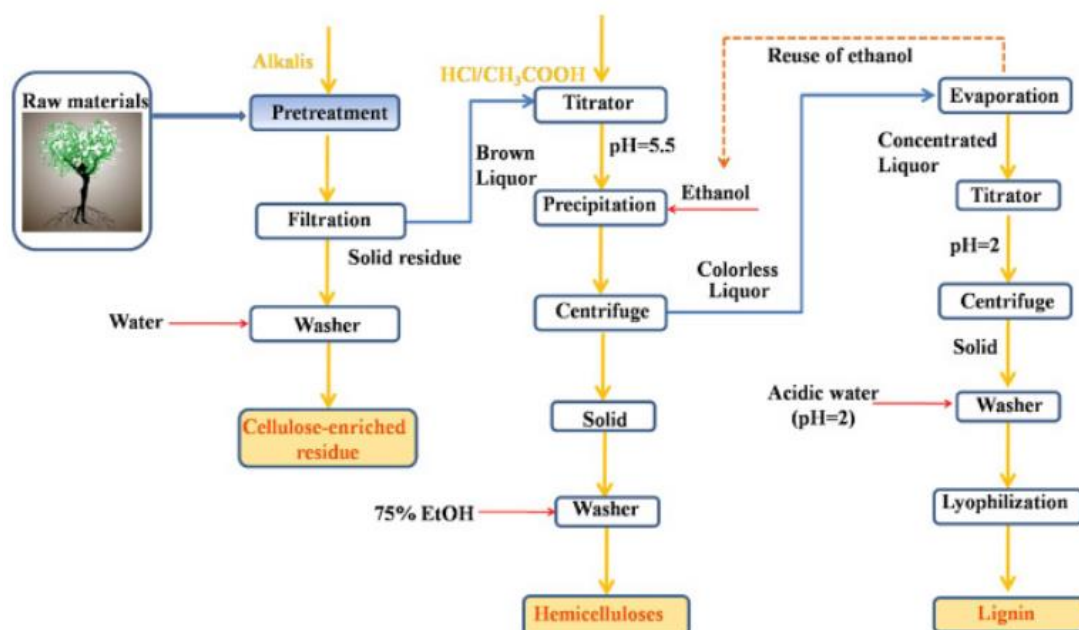


Figure-14: The conventional flow diagram of the pretreatment and fractionation of lignocellulosic biomass by an alkali-based process.

- Furthermore, alkali pretreatment is more favorable for low lignin content biomass such as herbaceous crops and agricultural residues and less productive for hardwoods.
- Selected precipitation approaches of hemicelluloses and lignin produced by alkaline pretreatment of biomass are summarized in **Figure-14**.
- Alkali pretreatment can be performed at room temperature for a few seconds up to several days. Sodium, potassium, calcium, and ammonium hydroxides are frequently used reagents for alkaline pretreatment and can improve the lignin removal efficiency.
- Pretreatment with sodium hydroxide (NaOH) is one of the most common methods and has been extensively studied in the bioconversion of lignocelluloses.
- NaOH treatment is very effective in increasing the digestibility of hardwood and agricultural residues with a low lignin content.
- Usually, alkaline pretreatment is carried out at low temperature and pressure, and it may be completed even at ambient conditions.
- The main drawback of this process is time; it might take hours or even days to complete the reaction.
- Temperature is an important factor that influences the performance of alkaline pretreatment.
- The reaction time could be saved through the increment of temperature.
- During lime [$\text{Ca}(\text{OH})_2$] pretreatment, some calcium is tainted into nonrecoverable salts or included in the biomass.
- Other alkaline pretreatment methods include calcium, potassium, sodium, barium, lithium, and ammonium hydroxides as reactants based on biomass category.
- Among these reactants, sodium hydroxide receives the most attention followed by lime, due to its advantage of being inexpensive and secure to use as well as being easily recoverable from water as an insoluble calcium carbonate by reaction with carbon dioxide.

- Further delignification of feedstocks can be enhanced by supplying surplus air or oxygen.
- We can compare alkaline pretreatment of feedstocks to Kraft pulping, where lignin was removed efficiently, thus improving the reactivity of the polysaccharides.
- Alkaline hydrolysis also effectively removes acetyl groups and uronic substitutions from hemicelluloses; thus the surface of hemicelluloses becomes more accessible to the hydrolytic enzymes.
- Furthermore the general results of alkaline pretreatment should be weighed against their influence on the ease of operation, cost of the downstream processes, and the trade-off between several costs, including operational, capital, and biomass prices.
- Hence, the cost-effective pretreatment of lignocellulosic biomass is a predominate technical bottleneck of cellulose conversion and multicomponent recovery'

Acid Pretreatment

- The primary objective through acid pretreatment is to solubilize the hemicellulosic portion of the LCB.
- Treatment can occur either at low-acid concentration and high temperature (dilute acid pretreatment) or high-acid concentration and low temperatures (concentrated acid pretreatment).
- While both dilute or concentrated acids could be used for pretreatment, using concentrated acids is less desirable because it produces more inhibiting compounds, such as HMF and furfural, than diluted acids.
- It also can cause greater corrosion and acid recovery problems compared to diluted acid, which increase the cost on a commercial scale.
- Diluted acid pretreatments have been studied for a variety of reactor types.
- The pretreatment can occur at high temperature (180 °C) for a short period of time, or a lower temperature (120°C) for a longer time period.
- CH₃COOH, HCl and H₂SO₄ pretreatments have been employed for improving the anaerobic digestion of lignocellulosic materials.
- Pretreatment with acid hydrolysis (HCl, H₂SO₄), can result in improvement of enzymatic hydrolysis of lignocellulosic biomass, to release fermentable sugars.
- Acid pretreatment results in the disruption of the van der Waals forces, hydrogen bonds and covalent bonds that hold together the biomass components, which consequently causes the solubilization of hemicellulose and the reduction of cellulose.
- The main reaction that occurs during acid pretreatment is the hydrolysis of hemicellulose, especially xylan, as glucomannan is more stable.
- Under such conditions, furfural and HMF generation can occur, because of dehydration of xylose galactose, mannose and glucose.
- Dilute acid hydrolysis pretreatment on the other hand can achieve high reaction rates and significantly improve cellulose hydrolysis.
- Lignin is hardly dissolved in most cases, but is disrupted to a high degree, thus leading to increased susceptibility of the cellulose to the enzymes.
- These pretreatments are more successful with usual concentration less than 4 wt%.
- Acid reagents, such as H₂SO₄, HCl, and CH₃COOH, at concentrations of 1, 2, and 4% (w/w) have been used for pretreatment.

- The most studied acid for pretreatment is diluted H_2SO_4 , which may in fact be the most common chemical pretreatment method.
- Experiments have shown the ability to remove nearly 100% of the hemicellulose with dilute sulfuric acid although it is not effective at dissolving lignin.
- It does disrupt lignin as well as increase the cellulose susceptibility for enzymatic hydrolysis.
- The conditions that are optimal for hemicellulose removal may not be optimal for the pretreatment for enzymatic hydrolysis.
- Studies testing various parameters of the pretreatment of olive tree biomass showed different process conditions for highest overall sugar yield, highest hemicellulose removal, and highest enzymatic hydrolysis yield.
- The major drawback of this method is formation of inhibitors. Compounds such as carboxylic acid, furfural, and phenolic compounds were formed, which may affect enzymatic hydrolysis and have a negative impact on microbial growth.
- Additional acids used for dilute acid pretreatment are the organic acids, such as fumaric or maleic acids. Initial studies show that less furfural was formed when using these organic acids, as opposed to sulfuric acid.
- However, as fewer studies have been performed with organic acids, it is unknown if they can achieve the same results as dilute sulfuric acid.
- A schematic diagram proposed for dilute acid pretreatment of biomass, Switchgrass (*Panicum Virgatum*) has been depicted in **Figure-15**.

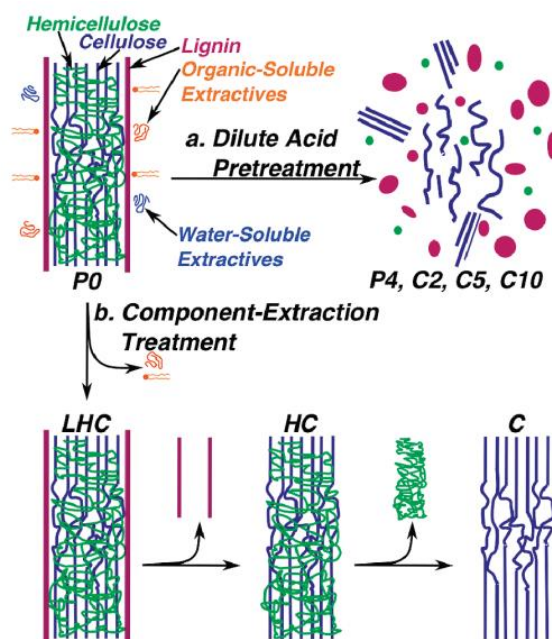


Figure-15: Schematic of (a) dilute acid pretreatment and (b) component extraction processes.

Ionic Liquids

- Ionic liquids (ILs) as cellulose solvent and ever since pretreatment of LCB by ILs has become a promising prospect.
- ILs are relatively new class of solvents with melting point $<100\text{ }^{\circ}\text{C}$ which are comprised of cations and anions.

- The cations, in general, are organic viz. imidazolium, pyridinium, aliphatic ammonium, alkylated phosphonium, and sulfonium ions, while the anions include both organic and inorganic ions.
- During the pretreatment process both the cations and anions play a significant role in solubilizing the cellulose and lignin. **Figure-16** represents the interruption of the intra- and intermolecular hydrogen bonding in cellulose by IL ions.
- Also, the cellulose dissolution increases in presence of electron-withdrawing groups in the alkyl chains of IL cations.

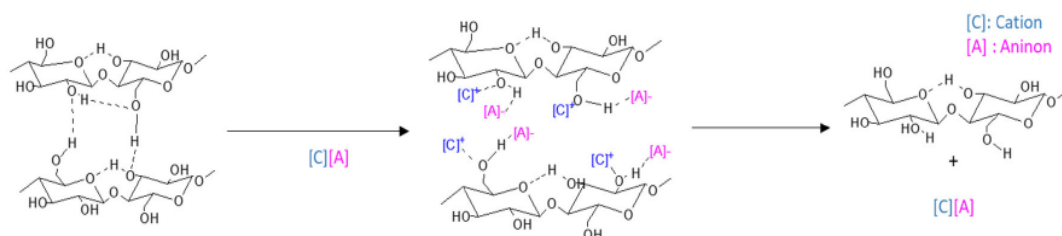


Figure-16: Interruption of intra- and intermolecular H-bonding in cellulose by ILs.

- Most of the ILs are recoverable and reusable.
- They possess the striking advantages of negligible vapor pressure, non-volatility, non-toxicity, large thermal, and chemical stability and most importantly the adjustable nature of their cations and anions on which the properties of the ILs depend.
- These are the reasons why ILs have often been described as green solvents.
- Several types of ILs include imidazolium-based $[(C_3N_2)X_n]^+$, pyridinium-based $[(C_5N)X_n]^+$, pyrrolidinium-based $[(C_4N)X_n]^+$, ammonium-based $[NX_4]^+$, phosphonium-based $[PX_4]^+$, sulfonium-based $[SO_3]^+$, and others such as cholin.
- Among these, imidazolium salts are the most commonly used ILs.
- Regardless of their distinctive chemical properties, ILs present the major shortcomings of being expensive and toxic to microorganisms and enzymes.
- Brandt-Talbot et al. (2017) investigated for the first time the application of the low-cost IL triethylammonium hydrogen sulfate for the pretreatment of *Miscanthus x giganteus* grass.
- They found up to 75% lignin and 100% hemicellulose solubilization in the IL solution and a yield of 77% glucose by enzymatic saccharification.
- Also the IL was reused four times with 99% recovery at each time.
- Nevertheless, further studies on these aspects with low-cost recovery technology and its toxicity to enzymes are still needed for economically-viable large-scale applications.

Organosolv Process

- In this process, LCBs are pretreated with organic solvents or their aqueous solutions that causes break down of the internal bonds between lignin and hemicellulose thereby leaving a relatively pure cellulose residue.
- During the process delignification and solubilization of hemicellulose lead to increase of pore volume and surface area of cellulose and enhances the accessibility of enzymatic hydrolysis and saccharification.

- A wide range of organic solvents such as, ethanol, methanol, acetone, organic acid, organic peracid, and ethylene glycol or their mixture with water have been employed to pretreat various LCBs.
- The process is often accompanied by the addition of a catalyst to either lower the pretreatment temperature or improve the delignification rate.
- Usually mineral acids (hydrochloric acid, sulfuric acid, phosphoric acid), bases (lime, sodium hydroxide, ammonia) and some salts are used as catalysts.
- Organosolv presents itself as an emerging pretreatment process owing to its inherent advantages such as, easy recovery of the solvents by distillation, recycles the solvents back to pretreatment and utilization of high-quality lignin isolated from this process as value-added byproducts for industrial applications.
- However, there also lies a few major disadvantages to the organosolv pretreatment.
- Most of the organic solvents are too expensive and need to be recovered as much as possible which is an energy-intensive process.
- In addition, high flammability and volatility of the organic solvents make the pretreatment to be carried out under especially controlled conditions.
- The organosolv pretreatment process for various feedstocks has been summarized in **Table-2**.

Table-2: Effect of organosolv pretreatment on different lignocellulosic substrates.

Raw material	Solvent	Catalyst	Condition	Results	References
Sorghum bagasse	25% Butanol	0.5% H ₂ SO ₄	200°C, 60 min	84.9% cellulose	Teramura et al., 2017
Barley straw	50% Acetone	0.5% H ₂ SO ₄	140°C, 20 min	83.9% glucose	Salapa et al., 2018
Sugarcane bagasse	60% Ethanol	0.025% FeCl ₃	160°C, 72 h	93.8% glucose	Zhang et al., 2018
Sugarcane bagasse	70% Glycerol	-	220°C, 120 min	94% cellulose	Sun F. F. et al., 2016
Wheat straw	25% Ethanol	1% H ₂ SO ₄	190°C, 60 min	80% glucose	Vergara et al., 2018
Wheat straw	50% Ethanol	0.35% H ₂ SO ₄	180°C, 40 min	89% cellulose	Salapa et al., 2017
Wheat straw	70% Glycerol	-	220°C, 180 min	70% cellulose	Sun F. F. et al., 2015
Corn stover	60% Ethanol	n-propylamine	140°C, 40 min	87.1% glucose	Tang et al., 2017b
Corn stalk	60% Ethanol	4% NaOH	110°C, 90 min	85% cellulose	Tang et al., 2017a
Rice straw	45% Ethanol	1% H ₂ SO ₄	180°C, 30 min	58.44% glucan	Asadi and Zilouei, 2017
Rice straw	65% Ethanol	1.1% H ₂ SO ₄	170°C, 60 min	70% cellulose	Sannigrahi et al., 2010
Bamboo	60% Ethanol	-	160°C, 60 min	67.2% glucose	Mou and Wu, 2017
Eucalyptus wood	56% Glycerol	-	200°C, 69 min	99% cellulose	Romani et al., 2016
Sweet sorghum	50% Ethanol	1% H ₂ SO ₄	140°C, 30 min	77% total sugar	Ostovareh et al., 2015

Deep Eutectic Solvents

- Pretreatment of LCB using deep eutectic solvents (DES) has attracted much interest in recent years.
- DESs are a new generation of ionic fluids comprising of two or three components, often interlinked through hydrogen bonding and form a eutectic mixture with a lower melting point than each individual components.
- They are usually liquids at temperature <100°C.
- DESs and ILs are much alike in terms of their physicochemical properties but their low-cost synthetic technology and biodegradability make them versatile alternatives to ILs .
- DESs are mostly derived by mixing a quaternary ammonium salt with a metal salt or hydrogen bond donor (HBD) which is capable of forming a complex with the halide ion of the quaternary ammonium salt.
- The decrease in melting point of the eutectic mixture is ascribed to the charge delocalization occurring between the halide ion and the hydrogen-donor moiety.

- DESs can be represented by the general formula

$$Cat^+ X^- zY \quad (1)$$

where Cat^+ is basically any ammonium, phosphonium or sulfonium cation, X^- is a Lewis base, commonly a halide anion, Y is Lewis or Brønsted acid, and z is the number of Y molecules that interact with the anion.

- The complexation between X^- and Y forms different anionic species.
- Being a biodegradable, non-toxic and cheap organic salt, cholin chloride (ChCl) are used in most of the DESs in combination with the low-risk HBDs like, urea, glycerol, carboxylic acids, and polyols.
- A study on the corn stover pretreatment by different DESs having the same halide salt i.e., ChCl found ChCl:Formic acid to be the ideal mixture for butanol fermentation.
- The findings implied that DES with acidic hydrogen donors could enhance the removal of lignin and hemicellulose more efficiently and are superior to ILs.
- The whole outline of DESs for biomass pretreatment and conversion has been depicted in the scheme demonstrated by **Figure-17**.

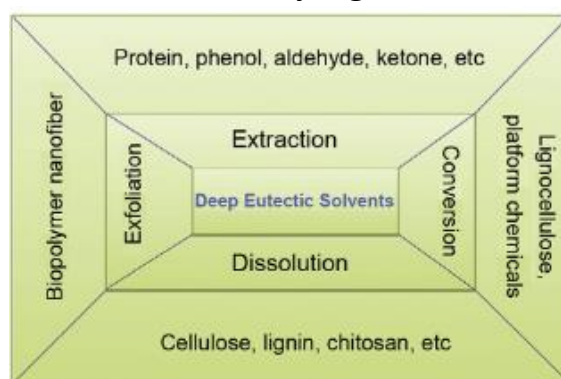


Figure-17: The whole outline of DESs for biomass pretreatment and conversion.

- Treatment of lignin with ILs required a high cost and a large amount of water.
- One-pot synthesis of biofuel production using ILs also needed the adjustment of pH and the dilution before saccharification/fermentation or after pretreatment.
- However, it was shown that one-pot production bio-ethanol can be achieved by using DESs via the couple of pretreatment, saccharification and fermentation as shown in the **Figure-18**.

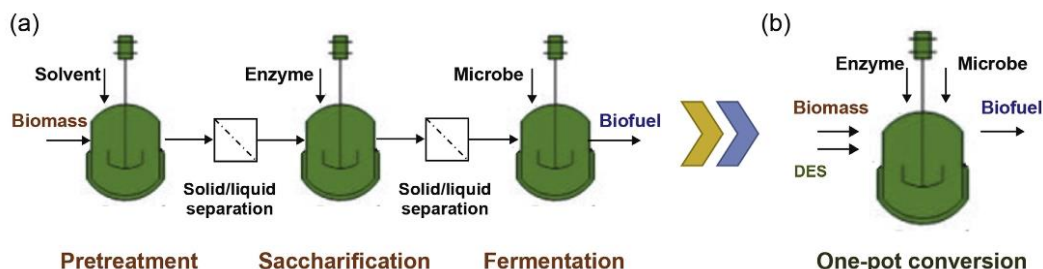


Figure-18: One-pot bioethanol production with biocompatible DESs.

○ Physicochemical (Thermochemical) Pretreatments

Steam Explosion

- ❖ Steam explosion is the most commonly employed and effective pretreatment method, which is typically a combination of both mechanical forces and chemical effects applied to LCBs.
- ❖ In this technique, the biomass is subjected to high-pressure saturated steam (0.69–4.83 MPa) at a temperature of 160–260 °C to let water molecules penetrate the substrate structure.
- ❖ The pressure is then suddenly reduced to let the water molecules escape in an explosive way.
- ❖ This rapid release of pressure causes explosion of the bulk LCB into splitted fibers.
- ❖ Besides, the high temperature and pressure enhance the breakdown of the glycosidic bonds in cellulose and hemicellulose and cleavage of hemicellulose-lignin bonds.
- ❖ During this treatment, the hydrolysis of hemicellulose into glucose and xylose monomers liberate acetic acid which catalyze the further hydrolysis of hemicelluloses, and; hence the process is also termed as autohydrolysis.
- ❖ Steam explosion has several advantages such as low environmental effect, limited chemicals use, high energy efficiency, no recycling costs and total sugar recovery compared with other pretreatment methods.
- ❖ It was found that almost 70% more energy is needed by the conventional mechanical methods than steam explosion to attain the same particle size reduction.
- ❖ This pretreatment is affected by several factors such as steam temperature, residence time, the size of biomass and moisture content.
- ❖ Steam explosion can be employed directly to milled LCB without employing any chemicals.

Ammonia Fiber Explosion (AFEX)

- ❖ In AFEX process, the LCB is heated with liquid ammonia (in 1:1 ratio) in a closed vessel at temperature around 60–100 °C under high pressure for 5–30 min, and then the pressure is suddenly released .
- ❖ The high pressure and given temperature causes swelling of lignocellulose and the rapid release of pressure disrupts the fibrous structure of biomass, reduces the crystallinity of cellulose and thereby improves the accessibility of enzyme.
- ❖ Optimization of AFEX pretreatment can be done by varying the four parameters including temperature, blowdown pressure, water loading and ammonia loading.
- ❖ Silvergrass (*Miscanthus spp.*) when presoaked in water prior to AFEX pretreatment showed up to 10% increase in glucan conversion indicating that moisture content plays a major role in AFEX.
- ❖ AFEX pretreatment partly removes the lignin and hemicellulose from lignocellulosic materials, but shows better enzymatic hydrolysis results at low enzyme loadings compared to other pretreatment processes.
- ❖ When oil palm empty fruit bunch fiber were pretreated with AFEX at 135 °C for 45min, the lignin-carbohydrate linkages changed along with some relocalization of lignin and resulted in 90% glucan conversion at lower enzyme loading of 13.8 FPUg⁻¹ glucan as compared to 25.5% conversion for untreated biomass.

- ❖ Therefore, AFEX pretreatment is more suitable for low lignin-content LCB such as agricultural wastes and herbaceous plants like switch grass, rice straw, corn stover etc.
- ❖ The main advantage of AFEX lies in the negligible formation of inhibitors as compared to other pretreatment methods.
- ❖ Nonetheless, ammonia should be recovered and recycled due to its high cost and volatility to reduce the overall operating cost and minimize environmental damage.
- ❖ The following **Figure-19** shows the fundamentals of ammonia fibre explosion:

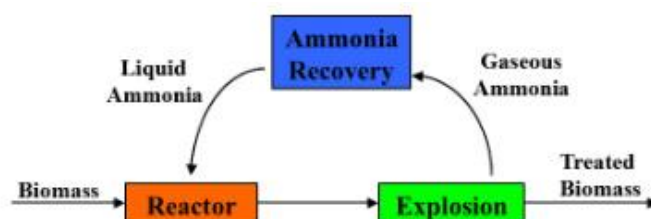


Figure-19: Ammonia fibre explosion.

CO₂ Explosion Pre-treatment

- ❖ A key drawback in the application of steam explosion pretreatment lies in its use of high thermal energy which is essential for the decomposition of the LCB.
- ❖ The AFEX pretreatments on the other hand use ammonia, which is highly corrosive with detrimental environmental effects, thereby limiting its scope.
- ❖ In this context, supercritical CO₂ explosion appears to be a viable alternative owing to its lower energy requirements and use of the greener alternative compared to ammonia.
- ❖ Because of its characteristics of mass transfer of a “gaslike” with a “liquidlike” solvating power, supercritical CO₂ can diffuse through interspaces like gas and dissolve materials like liquid.
- ❖ Under high pressure the CO₂ molecules penetrate into the biomass and shattered the higher level of structures comprising hemicellulose and lignin.
- ❖ Once dissolved in water, CO₂ forms carbonic acid, which catalyzes the hydrolysis of hemicellulose.
- ❖ This is the reason why this pretreatment process is not satisfactory to biomass with no moisture content.
- ❖ On the other hand, when the pressurized gas is released, it breaks the compact matrix structure of biomass thereby improving the accessibility of cellulose fibers.
- ❖ The penetrating rate of the CO₂ molecules into the cellulosic pores increases with pressure and leads to high glucose yield.
- ❖ Also, it was reported that addition of water-ethanol as co-solvents during supercritical CO₂ explosion, pretreatment can significantly remove the lignin and enhance enzymatic hydrolysis of corn stover.
- ❖ Benazzi et al. (2013) reported a single-step process for the hydrolysis of sugarcane bagasse using supercritical CO₂ and obtained 60% yield of fermentable sugars.
- ❖ Soybean hull pretreated with supercritical CO₂ showed a high glucose yield of 97% after enzymatic hydrolysis.

- ❖ This high glucose yield was obtained under optimal conditions of 8 MPa pretreatment pressure and 130 °C temperature for 30 min.
- ❖ Thus the pretreatment pressure and temperature also plays a significant role in supercritical-CO₂ treatment.

Liquid Hot Water (LHW)

- ❖ LHW pretreatment is very much alike to steam explosion but as the name suggests, LHW uses water at elevated temperature (170–230 °C) and pressure (up to 5MPa) in place of steam.
- ❖ Unlike steam explosion rapid release of pressure is not required in LHW and the application of pressure is only to prevent evaporation of water.
- ❖ LHW hydrolyzes hemicellulose by liberating its acetyl groups and removes lignin thereby making the cellulose fibers more exposable.
- ❖ The detached hemicellulose remains in the liquid fraction of the pretreated slurry and formation of monomeric sugars during the process is minimal.
- ❖ However, in order to avoid the sugar degradation and inhibitors formation, the LHW pretreatment is carried out at controlled pH between 4 and 7.
- ❖ Furthermore, biomass size has no influence since the particles are broken down during the treatment which makes the process more striking for large scale.
- ❖ However, the process is very energy intensive due to a large amount of water involved.

○ Biological Pre-treatment

Biological pretreatment is a low cost and eco-friendly technique to treat LCB prior to enzymatic saccharification. This technique is promising as there is no inhibitor formation during the process, requires lesser energy and is eco-friendly. Through this method, lignin degrading bacteria or fungi, as whole cell or enzymes, are used to pretreat LCB. The enzymes used in degradation of lignin are laccases, lignin peroxidase, manganese peroxidase, and versatile peroxidase. Fungi are the best suited for such applications as they are capable of degrading cellulose, hemicelluloses, and lignin. Biological pretreatment is not only used for lignin removal, but also for removal of specific components such as antimicrobial substances. White-rot, brown-rot and soft-rot fungi are used for degradation of lignin and hemicelluloses present in LCB. However, mostly white-rot fungi are involved in biological pretreatment due to high sugar yield associated with enzymatic saccharification. Some white-rot fungi can simultaneously degrade lignin and polysaccharides, resulting in the loss of carbohydrates, while other white-rot fungi can selectively degrade lignin.

There are two extracellular enzymatic systems involved in microorganisms, one hydrolytic and another ligninolytic system. Hydrolytic system is responsible for degradation of cellulose and hemicelluloses, while ligninolytic system depolymerizes the lignin. Lignin can be degraded by enzymes produced by various organisms, among which white-rot fungus has been found the most effective. Fungi are usually isolated from the soil, living plants, or agricultural waste materials.

Whole Cell Pre-treatment

- White-rot fungi are generally used as whole cell microorganism since it is less potent for the degradation of the cellulosic fraction of the LCB.

- The ligninolytic system in these fungi secretes one or more extracellular enzymes which are responsible for the degradation of lignin (aromatic polymer) and aliphatic fragments.
- White-rot fungi commonly employed for ligninolytic pretreatment are *Phanerochaete chrysosporium*, *Ceriporiopsis subvermisporea*, *Ceriporia lacerata*, *Cyathus stercolerus*, *Pycnoporus cinnabarinus*, *Pleurotus ostreatus*, *Phlebia subserialis*, *Pleurotus streatus*, *Postia placenta*, *Gloeophyllum trabeum*, and *Echindodontium taxodii*.
- The above fungi are capable of efficiently metabolizing lignin in a variety of LCBs.
- Among these, *Phanerochaete chrysosporium* is a model organism for lignin degradation.
- However, white-rot fungi may face challenges in lignin degradation due to presence of carbon-carbon bonds within the large lignin polymers.
- **Table-3** summarizes the different biological pretreatment conditions that have been used in the recent past for treatment of various biomass feedstocks using whole cell biocatalyst.
- The major disadvantages of biological pretreatment methods are low efficiency and long residence periods, when used in isolation.

Table-3: Biological Pretreatment Conditions for Various Feed stocks.

Feedstock	Microorganism	Process condition	Effect on biomass	References
Sugarcane bagasse	<i>Ceriporiopsis subvermisporea</i>	27 ± 2°C for 60 days	47% sugar was recovered as sugar-rich syrup	Machado and Ferraz, 2017
Sawdust	<i>Pleurotus pulmonarius</i>	28°C for 30 days	Sugar concentration increased from 2.5l mol/mL to 48.0l mol/mL.	Castoldi et al., 2014
Wheat straw	<i>Ceriporiopsis subvermisporea</i>	7 weeks of solid state fermentation incubated at 24°C	<i>in-vitro</i> gas production of 297.0 ml g ⁻¹	Nayan et al., 2018
Paddy straw	<i>Pleurotus florida</i>	25–29°C for 28 days.	Maximum saccharification efficiency up to 75.3%	Manickam et al., 2018
Straw	Fungal consortium	Incubated at 30 and 55°C for 6 days	7-fold increase in hydrolysis	Taha et al., 2015
Corn stover	Fungal consortium	Pretreatment for 42 days with fungi	43.8% lignin removal and 7-fold increase in hydrolysis	Song et al., 2013
Corn stalks	<i>Irpex lacteus</i>	28 days incubation	82% hydrolysis yield was observed	Du et al., 2011
Bamboo clums	<i>Punctularia sp. TUFC20056</i>		50% of lignin removed	Suhara et al., 2012
Corn stover	<i>Ceriporiopsis subvermisporea</i>	Solid state fermentation at 28°C for 42 days	57–67% glucose yield increase	Wan and Li, 2010
Rice straw	<i>Pholiota adiposa</i>	Saccharification for 48 h	Releases 716 mg-sugar g-substrate ⁻¹	Jagtap et al., 2013a
Aspen biomass	<i>Armillaria gemina</i> SKU2114	48 h of hydrolysis	Maximum saccharification yield of 62%	Jagtap et al., 2013b

Enzymatic Pretreatment

- Lignin is the most abundant aromatic polymer consisting of phenolic and non-phenolic compounds.
- Some fungi, bacteria, and insects are capable of producing enzymes which digest lignin.
- There are two families of ligninolytic enzymes which play an important role in enzymatic degradation: phenol oxidase (laccase) (Lac) and peroxidases (lignin peroxidase (LiP), versatile peroxidase (VP), and manganese peroxidase (MnPP)).
- These are heme-containing glycoproteins which require hydrogen peroxide (H₂O₂) as oxidant.

- The role of other enzymes has not yet been fully elucidated including glyoxal oxidase (GLOX; EC 1.2.3.5), glucose oxidase (EC 1.1.3.4), cellobiose dehydrogenase (CDH; EC 1.1.99.18) oxidoreductase and methanol.
- There are different microorganisms i.e. bacteria and fungi, which produce cellulolytic, hemicellulolytic and ligninolytic enzymes.
- Lignin degrading enzymes are directly employed to pretreat the biomass. These are employed in a group of one or more.
- **Table-4** shows the effect of enzymatic pretreatment on different lignocellulosic feedstocks.

Table-4: Effect Of Enzymatic Pretreatment On Different Lignocelluloses

Enzyme	Source strain	Substrate	Conditions	Effect of biomass	References
Laccase and peroxidase	<i>Polyporus brumalis_BRFM985</i>	Wheat straw	21 days old fungal pretreated wheat straw	39% loss of lignin	Zhou X. et al., 2017
Laccase	<i>Tyromyces chioneus</i>	Rice straw and willow	Simultaneous pretreatment and saccharification (SPS)	Removal of 49.8% and 32.6% of phenolic contents from the soaked rice straw and willow, respectively and saccharification yield of up to 74.2% and 63.6% for rice straw and willow, respectively	Dhiman et al., 2014
Laccase	<i>Trametes versicolor</i>	Wheat straw	Treated with dilute sulphuric acid	Released higher glucose concentration of up to 2.3 g L ⁻¹ as compared to control	Heap et al., 2014
Laccase	<i>Tyromyces chioneus</i>	Rice straw	in a rotary shaker 150 rpm	Saccharification yield was observed up to 74.2%	Dhiman et al., 2014
Laccase	<i>Myceliophthora thermophila</i>	Wood	45°C for 72 h in thermostate shaker 170 rpm	50% of lignin removed	Rico et al., 2014

- The following **Figure-20** (a) and (b) gives basic understanding of the enzyme pretreatment.

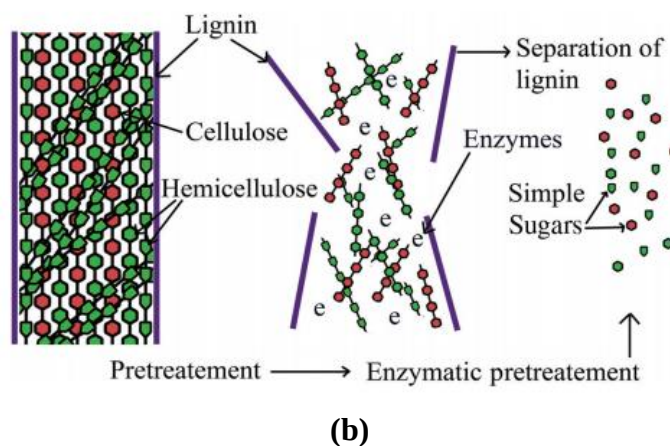
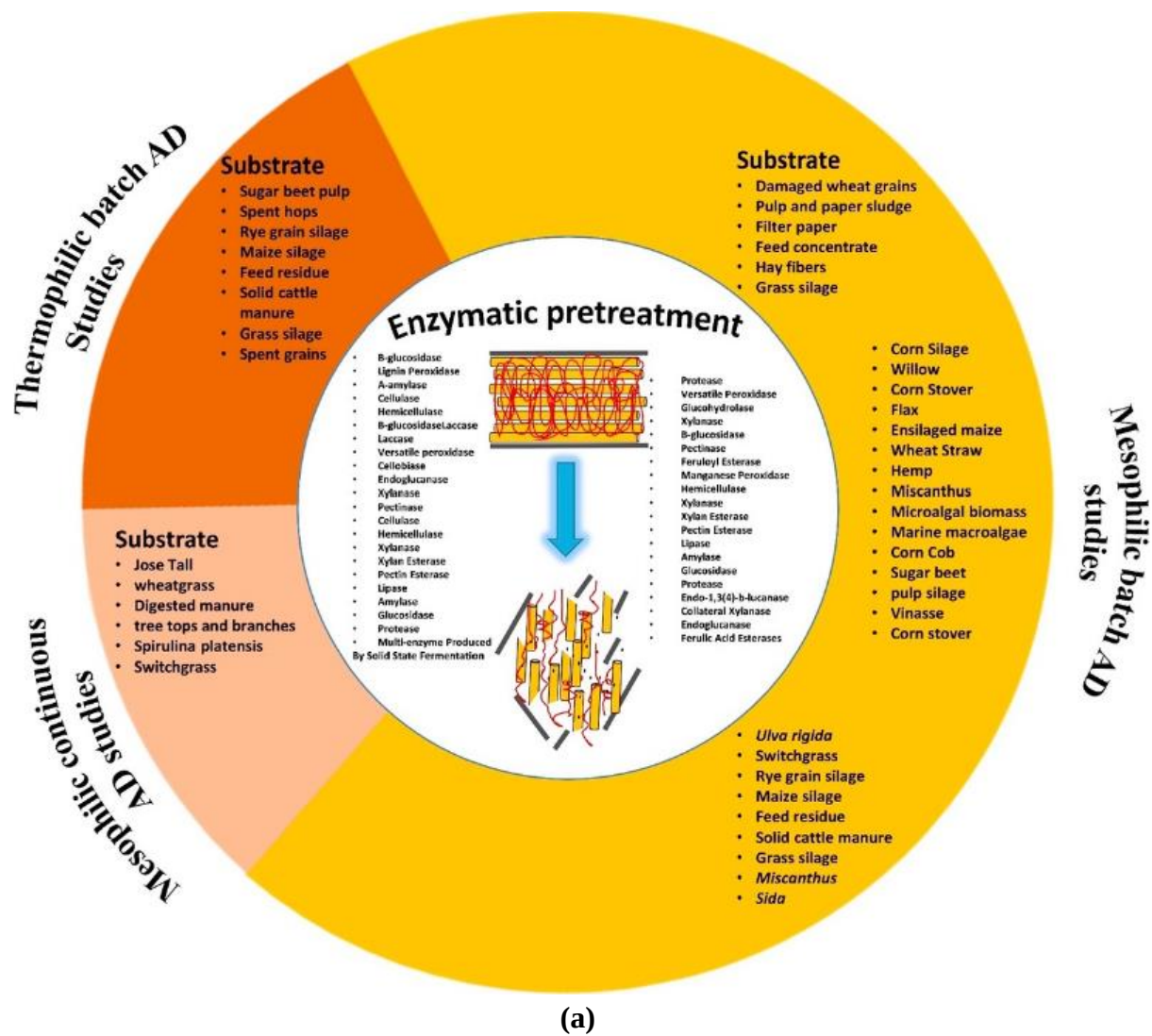


Figure-20: Enzyme pretreatment LCB.

○ **Lignin Peroxidase (LiP)**

- ❖ Lignin peroxidase (LiP) (EC 1.11.1.4) was first discovered in *Phanerochaete chrysosporium* grown in nitrogen-limited medium.

- ❖ LiP is H_2O_2 dependent glycoprotein, which contains heme and has a molecular mass ranges from 35 to 40 kDa.
- ❖ LiP has been reported to produce by white-rot fungi like *Phlebia flavido-alba*, *Bjerkandera* sp. strains BOS55, *Trametes trogii*, *Phlebia tremesllosa*, *Gloeophyllum trabeum*, *Trametes versicolor*, *Phanerochaete chrysosporium*, etc.
- **Laccases (Lac)**
 - ❖ Laccases (Lac) (EC 1.10.3.2) or benzenediol oxygen oxidoreductase is a multicopper enzyme that belongs to a group of blue copper containing oxidase.
 - ❖ White-rot fungi produced laccase, catalyzes the oxidation of aromatic amines and phenolic compounds such as phenolic substructure of lignin.
 - ❖ Lac is glycoprotein and its molecular weight ranges between 60 and 80 kDa.
 - ❖ It is also able to oxidize non-phenolic substructure of lignin in the presence of low molecular weight compounds hydroxyl benzotriazole.
 - ❖ Laccase producing microorganisms are both fungi (*Trametes versicolor*, *Tremetes trogii*, *Phlebia floridensis*), and bacteria (*Citrobacter* spp., *Straphylococcus saprophlticus*, *Bacillus subtilis*).
 - ❖ Lac enzyme plays an important role in lignin degradation and modification processes which increase the yield of both hydrolysis and fermentation process.
 - ❖ Laccase is an important enzyme since it oxidizes both toxic and non-toxic substrates.
 - ❖ It is used in textile, food processing, wood processing, pharmaceutical, and chemical industries.
 - ❖ Laccase enzyme is very specific, ecologically sustainable and a proficient catalyst.
 - ❖ Several bacterial and fungal laccases have been used for detoxification of various agro-waste (pretreated and unpretreated) feedstocks in the presence of mediator system.
- **Manganese Peroxidase (MnP)**
 - ❖ Manganese peroxidase (MnP) (EC 1.11.1.13), Mn (II) hydrogen peroxide oxidoreductase, catalyzes Mn dependent reactions.
 - ❖ Similar to LiP, MnP is also heme glycoprotein.
 - ❖ Major difference in MnP and LiP is: LiP generally oxidizes non-phenolic lignin, while MnP oxidizes phenolic ring of lignin and plays an important role in the initial stage of lignin degradation.
 - ❖ Manganese peroxidase oxidizes phenolic and non-phenolic lignin units through lipid peroxidation reactions and forms various phenolic compounds i.e., 3-ethylthiazoline-6-sulfonate, 2, 6-dimethyloxyphenol syringol, guaiacol, and non-phenol compound i.e., alcohol.
 - ❖ It oxidizes Mn^{2+} to Mn^{3+} which further oxidizes phenol rings to phenoxy radicals leading to decomposition of compounds.
- **Versatile Peroxidase (VP)**
 - ❖ Versatile peroxidase (VP) (EC 1.11.1.16) oxidizes phenolic and non-phenolic aromatic compounds.
 - ❖ VP enzymes have the catalytic activities of both MnP and LiP, and are able to oxidize Mn^{2+} like MnP as well as high-redox potential non-phenolic compounds like LiP.

- ❖ VP is employed together with MnP, LiP and other microbial peroxidases in degradation of non-phenolic aromatic compounds such as veratrylglycerol β -guaiacyl ether to veratraldehyde. VP also oxidize Mn^{2+} to Mn^{3+} , veratyl alcohol to veratraldehyde and p-dimethoxybenzene to p-benzoquinone.
- ❖ VP comprises the ligninolytic heme peroxidase gene family of *Pleutotus ostreatus*.
- ❖ It is found in various *Bjerkandera species* and *Pleurotus species*.
- ❖ It catalyzes oxidation of wide range of substrates from plant peroxidase hydroquinones, substituted phenols to bulky recalcitrant lignin directly, without redox mediators.
- ❖ It is used in textile, bleaching, paper and pulp industries, production of biofuels, bioremediation of xenobiotic compounds, degradation of endocrine disrupting chemicals.

➤ ENZYME (BIOLOGICAL) HYDROLYSIS

- ✓ *Enzymatic hydrolysis* is the process in which cellulases are added to *hydrolyze* pretreated lignocellulosic biomass into fermentable sugars.
- ✓ Enzymatic hydrolysis is the breakdown of a compound in presence of enzymes following its reaction with water.
- ✓ It has been extensively used in food and power industries.
- ✓ Power industries use enzyme hydrolysis to help provide renewable energy, as with cellulosic biodiesel, ethanol, bioethanol etc.
- ✓ The rate of *enzymatic hydrolysis* is also influenced by *enzyme* sources and the proportion of different *enzyme* components.
- ✓ **Ionic Liquid Enzyme Hydrolysis**
 - Biofuels and -chemicals can be produced from carbohydrates in lignocellulosic biomass.
 - For an efficient total enzymatic hydrolysis of the plant cell wall polysaccharides, a pretreatment step is required.
 - Ionic liquids (ILs) such as [BMIM]Cl, [BMIM]BF₄ have recently gained considerable interest as solvents for cellulose and lignocellulosic biomass and pretreatment of lignocellulose with ILs is currently an extensively studied concept.
 - However, the applicability of ILs in an integrated process, in which enzymatic hydrolysis is done in the same vessel as the IL pretreatment without IL removal and substrate washing between the process steps, suffers from the fact that cellulose-dissolving ILs severely inactivate the cellulases used to catalyse the polysaccharide hydrolysis.
 - Impressive advances have recently been made in discovering and developing cellulases and other glycosyl hydrolases with increased IL-tolerance.
 - Different cellulase stabilisation techniques and the design of enzyme-friendly cellulose-dissolving ILs are also available.
 - In light of the recent developments, the integrated enzymatic hydrolysis of polysaccharides in the presence of ILs may well prove to be a potential route for utilizing lignocellulosic biomass as feedstock in biofuel and -chemical production.

- The following **Figure-3** briefly explains the process of hydrolysis.

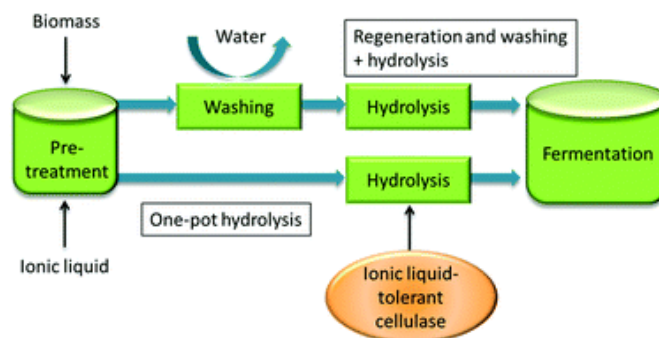


Figure-3: Ionic Liquids (ILs) enzyme hydrolysis of lignocellulosics.

- The main components of lignocellulosics are cellulose, hemicelluloses and lignin, in various ratios depending on the source of biomass.
- Together these polymers form a complex matrix, which is highly recalcitrant towards depolymerization.

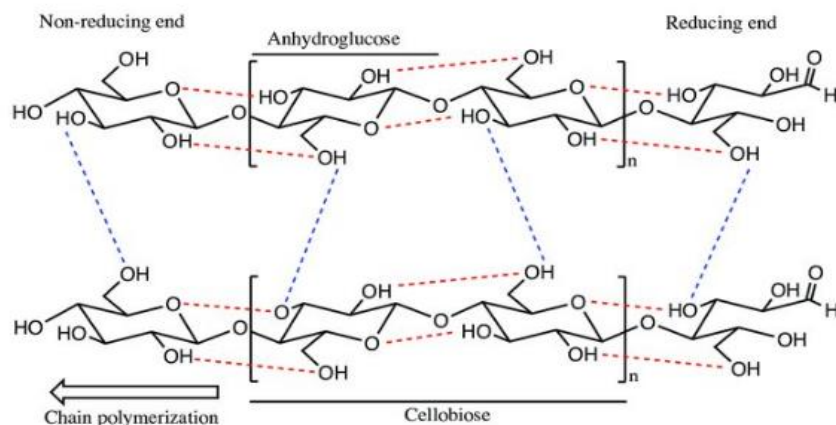


Figure-5: Cellulose structure with the intra--and inter--chain hydrogen bonds.

- Cellulose and hemicelluloses are polysaccharides.
- Cellulose is a linear homopolymer consisting of anhydroglucose units linked together by 1,4-β-glycosidic bonds.
- The anhydroglucose units are distorted 180° to each other so that the smallest repeating unit in the cellulose chain is the anhydrocellobiose unit (**Figure-4**).